

# CHAPTER 14

## Semiconductor Electronics

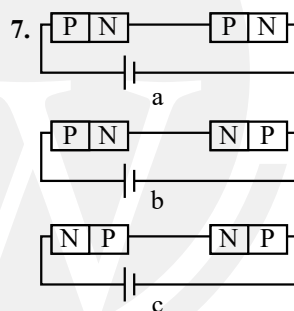
### Energy Band in Solids and Basics of Semiconductor

- The solids which have the negative temperature coefficient of resistance are: (2020)
  - Insulators only
  - Semiconductors only
  - Insulators and semiconductors
  - Metals

### Intrinsic and Extrinsic Semiconductor (P and N types)

- A p-type extrinsic semiconductor is obtained when Germanium is doped with: (2023-Manipur)
  - Antimony
  - Phosphorous
  - Arsenic
  - Boron
- The electron concentration in an n-type semiconductor is the same as hole concentration in a p-type semiconductor. An external field (electric) is applied across each of them. Compare the currents in them. (2021)
  - current in p-type > current in n-type.
  - current in n-type > current in p-type.
  - No current will flow in p-type, current will only flow in n-type.
  - current in n-type = current in p-type.
- An intrinsic semiconductor is converted into n-type extrinsic semiconductor by doping it with (2020-Covid)
  - Aluminium
  - Silver
  - Germanium
  - Phosphorous
- For a p-type semiconductor, which of the following statements is true? (2019)
  - Electrons are the majority carriers and trivalent atoms are the dopants.
  - Holes are the majority carriers and trivalent atoms are the dopants.
  - Holes are the majority carriers and pentavalent atoms are the dopants.
  - Electrons are the majority carriers and pentavalent atoms are the dopants.
- In a n-type semiconductor, which of the following statement is true? (2013)
  - Holes are majority carriers and trivalent atoms are dopants
  - Electrons are majority carriers and trivalent atoms are dopants
  - Electron are minority carriers and pentavalent atoms are dopants
  - Holes are minority carriers and pentavalent atoms are dopants

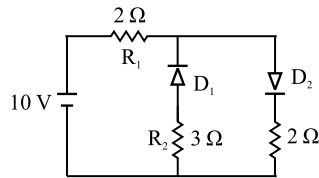
### P-N Junction Diode (forward and reverse bias), Diffusion and Drift Current



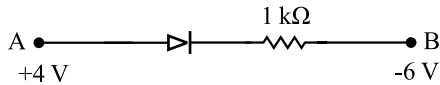
In the given circuits (a), (b) and (c), the potential drop across the two p-n junctions are equal in: (2022)

- Both circuits (a) and (c)
  - Circuit (a) only
  - Circuit (b) only
  - Circuit (c) only
- Out of the following which one is a forward biased diode? (2020-Covid)
    - $\frac{2V}{-}$   $\frac{5V}{+}$
    - $\frac{-2V}{-}$   $\frac{+2V}{+}$
    - $\frac{0V}{-}$   $\frac{-3V}{+}$
    - $\frac{-4V}{-}$   $\frac{-2V}{+}$
  - The increase in the width of the depletion region in a p-n junction diode is due to : (2020)
    - Reverse bias only
    - Both forward bias and reverse bias
    - Increase in forward current
    - Forward bias only
  - Which one of the following represents forward bias diode? (2017-Delhi, 2006)
    - $\frac{-4V}{-}$   $\frac{-3V}{+}$
    - $\frac{-2V}{-}$   $\frac{+2V}{+}$
    - $\frac{3V}{-}$   $\frac{5V}{+}$
    - $\frac{0V}{-}$   $\frac{-2V}{+}$

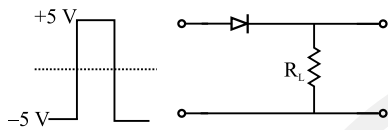
11. The given circuit has two ideal diodes connected as shown in the figure below. The current flowing through the resistance  $R_1$  will be: (2016-II)



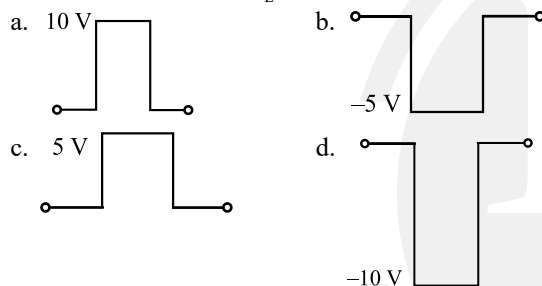
- a. 1.43 A    b. 3.13 A    c. 2.5 A    d. 10.0 A
12. Consider the junction diode as ideal. The value of current flowing through AB is: (2016-I)



- a. 0 A    b.  $10^{-2}$  A    c.  $10^{-1}$  A    d.  $10^{-3}$  A
13. If in a p-n junction, a square input signal of 10 V is applied, as shown



then the output across  $R_L$  will be: (2015)



### Characteristics of P-N Junction

14. In a p-n junction diode, change in temperature due to heating (2018)

- a. Does not affect resistance of p-n junction  
b. Affects only forward resistance  
c. Affects only reverse resistance  
d. Affects the overall V-I characteristics of p-n junction

15. The barrier potential of a p-n junction depends on: (2014)

- A. Type of semiconductor material  
B. Amount of doping  
C. Temperature

Which one of the following is correct?

- a. A and B only    b. B only  
c. B and C only    d. A, B and C

### Diode as a Rectifier

16. A full wave rectifier circuit consists of two p-n junction diodes, a centre-tapped transformer, capacitor and a load resistance. Which of these components remove the ac ripple from the rectified output (2023)

- a. Load resistance    b. A centre-tapped transformer  
c. p-n junction diodes    d. Capacitor

17. In half wave rectification, if the input frequency is 60 Hz, then the output frequency would be: (2022)

- a. 120 Hz    b. 0    c. 30 Hz    d. 60 Hz

### Zener Diode

18. Given below are two statements: (2023)

**Statement I:** Photovoltaic devices can convert optical radiation into electricity.

**Statement II:** Zener diode is designed to operate under reverse bias in breakdown region.

In the light of the above statements, choose the most appropriate answer from the options given below:

- a. Statement I is incorrect but Statement II is correct.  
b. Both Statement I and Statement II are correct.  
c. Both Statement I and Statement II are incorrect.  
d. Statement I is correct but Statement II is incorrect.

19. The incorrect statement about the property of a Zener diode is: (2022 Re)

- a. p and n regions of zener diode are heavily doped  
b. Zener voltage remains constant at breakdown  
c. It is designed to operate under reverse bias  
d. Depletion region formed is very wide

20. Consider the following statements (A) and (B) and identify the correct answer. (2021)

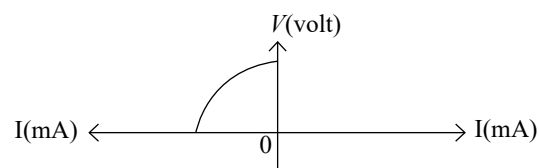
(A) A zener diode is Connected in reverse bias, when used as a voltage regulator.

(B) The potential barrier of p-n junction lies between 0.1 V to 0.3 V.

- a. (A) and (B) both are incorrect.  
b. (A) is correct and (B) is incorrect.  
c. (A) is incorrect but (B) is correct.  
d. (A) and (B) both are correct.

### Optoelectronic Devices

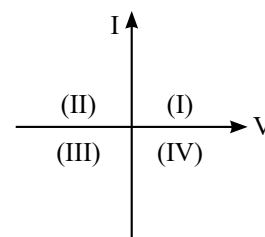
- 21.



The I-V characteristics shown above are exhibited by a (2024 Re)

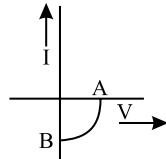
- a. Light emitting diode    b. Zener diode  
c. Photodiode    d. Solar cell

22. Consider the following statement A and B and identify the correct answer: (2024)



- A. For a solar-cell, the I-V characteristics lies in the IV quadrant of the given graph.  
 B. In a reverse biased *pn* junction diode, the current measured in (A), is due to majority charge carriers.  
 a. Both A and B are correct.  
 b. Both A and B are incorrect.  
 c. A is correct but B is incorrect.  
 d. A is incorrect but B is correct.

23. The given graph represents V-I characteristic for a semiconductor device. (2014)

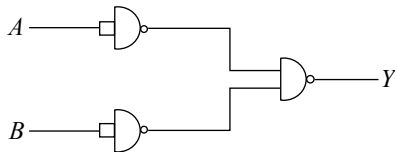


Which of the following statement is correct?

- a. It is V-I characteristic for solar cell where point A represents open circuit voltage and point B short circuit current  
 b. It is for a solar cell and points A and B represent open circuit voltage and current, respectively  
 c. It is for a photo diode and points A and B represent open circuit voltage and current, respectively  
 d. It is for a LED and points A and B represents open circuit voltage and short circuit current respectively

### Digital Electronics and Logic Gates

24. When the output of an OR gate is applied as input to a NOT gate, then the combination acts as a (2024 Re)  
 a. NAND gate                      b. NOR gate  
 c. AND gate                        d. OR gate
25. The output *Y* for the inputs *A* and *B* of the given logic circuit is: (2024 Re)

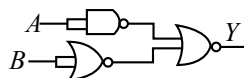


- a.  $A \cdot B$       b.  $\bar{A} \cdot \bar{B}$       c.  $A + B$       d.  $\bar{A} + \bar{B}$
26. A logic circuit provides the output *Y* as per the following truth table: (2024)

| A | B | Y |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

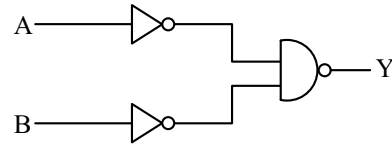
The expression for the output *Y* is:

- a.  $\bar{B}$       b.  $B$       c.  $A\bar{B} + \bar{A}B$       d.  $A\bar{B} + \bar{A}B$
27. The output (*Y*) of the given logic gate is similar to the output of an/a: (2024)



- a. OR gate                      b. AND gate  
 c. NAND gate                  d. NOR gate

28. For the following logic circuit, the truth table is: (2023)



| A | B | Y |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

a.

| A | B | Y |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

b.

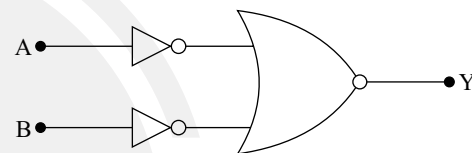
| A | B | Y |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

c.

| A | B | Y |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

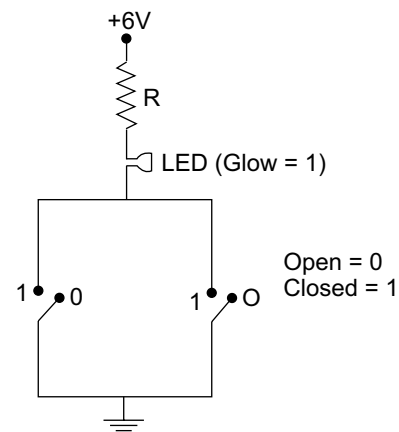
d.

29. The given circuit is equivalent to: (2023-Manipur)

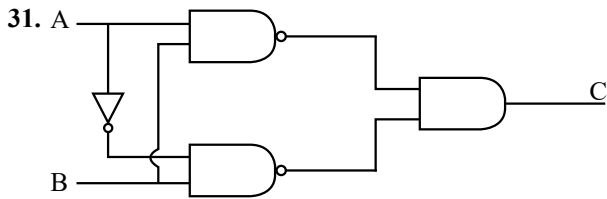


- a.
- 
- b.
- 
- c.
- 
- d.
- 

30. Identify the equivalent logic gate represented by the given circuit: (2022 Re)



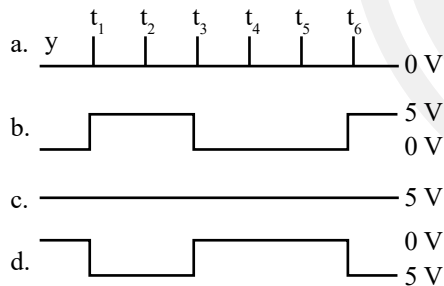
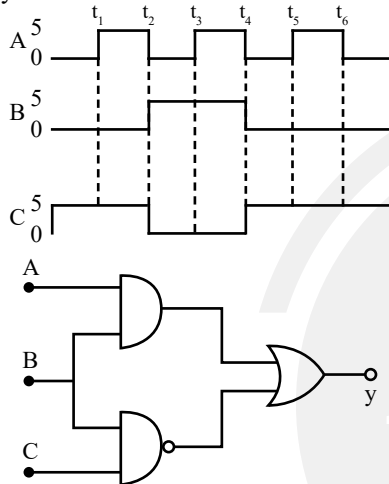
- a. NAND      b. OR      c. NOR      d. AND



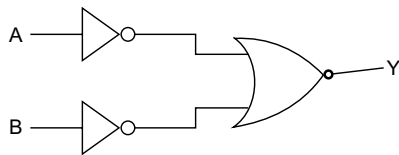
The truth table for the given logic circuit is: (2022)

| a. | A | B | C | b. | A | B | C | c. | A | B | C | d. | A | B | C |
|----|---|---|---|----|---|---|---|----|---|---|---|----|---|---|---|
|    | 0 | 0 | 0 |    | 0 | 0 | 0 |    | 0 | 0 | 1 |    | 0 | 0 | 1 |
|    | 0 | 1 | 1 |    | 0 | 1 | 1 |    | 0 | 1 | 0 |    | 0 | 1 | 0 |
|    | 1 | 0 | 0 |    | 1 | 0 | 1 |    | 1 | 0 | 0 |    | 1 | 0 | 1 |
|    | 1 | 1 | 1 |    | 1 | 1 | 0 |    | 1 | 1 | 1 |    | 1 | 1 | 0 |

32. For the given circuit, the input digital signals are applied at the terminals A, B and C. What would be the output at the terminal y? (2021)



33. For the logic circuit shown, the truth table is: (2020)



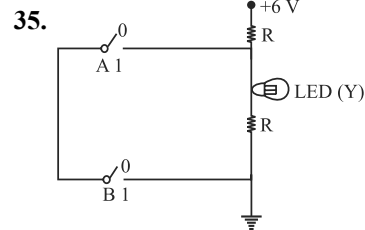
| a. | A | B | Y | b. | A | B | Y |
|----|---|---|---|----|---|---|---|
|    | 0 | 0 | 0 |    | 0 | 0 | 1 |
|    | 0 | 1 | 1 |    | 0 | 1 | 1 |
|    | 1 | 0 | 1 |    | 1 | 0 | 1 |
|    | 1 | 1 | 1 |    | 1 | 1 | 0 |

| c. | A | B | Y | d. | A | B | Y |
|----|---|---|---|----|---|---|---|
|    | 0 | 0 | 1 |    | 0 | 0 | 0 |
|    | 0 | 1 | 0 |    | 0 | 1 | 0 |
|    | 1 | 0 | 0 |    | 1 | 0 | 0 |
|    | 1 | 1 | 0 |    | 1 | 1 | 1 |

34. Which of the following gate is called universal gate? (2020-Covid)

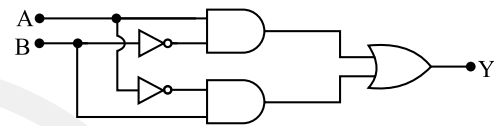
a. AND gate  
b. NAND gate  
c. NOT gate  
d. OR gate



The correct Boolean operation represented by the circuit diagram drawn is: (2019)

a. AND b. OR c. NAND d. NOR

36. In the combination of the following gates the output Y can be written in terms of inputs A and B as (2018)



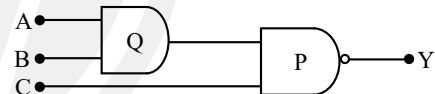
a.  $\overline{A.B} + A.B$   
b.  $A.\overline{B} + \overline{A}.B$   
c.  $\overline{A.B}$   
d.  $\overline{A + B}$

37. The given electrical network is equivalent to: (2017-Delhi)



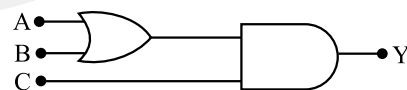
a. OR gate  
b. NOR gate  
c. NOT gate  
d. AND gate

38. What is the output Y in the following circuit, when all the three inputs A, B, C are first 0 and then 1? (2016 - II)



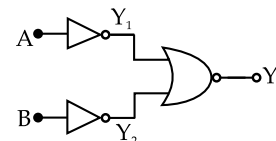
a. 1, 0  
b. 1, 1  
c. 0, 1  
d. 0, 0

39. To get output 1 for the following circuit, the correct choice for the input is: (2016-I, 2012 Mains, 2010 Pre)



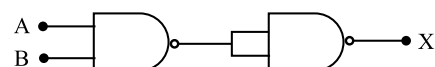
a. A = 0, B = 1, C = 0  
b. A = 1, B = 0, C = 0  
c. A = 1, B = 1, C = 0  
d. A = 1, B = 0, C = 1

40. Which logic gate is represented by the following combination of logic gates? (2015)



a. NAND b. AND c. NOR d. OR

41. The output (X) of the logic circuit shown in figure will be: (2013)



a.  $X = \overline{A + B}$   
b.  $X = \overline{\overline{A}.B}$   
c.  $X = \overline{A.B}$   
d.  $X = A.B$

**Answer Key**

- |         |         |         |         |         |         |         |         |         |         |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (c)  | 2. (d)  | 3. (b)  | 4. (d)  | 5. (b)  | 6. (d)  | 7. (a)  | 8. (c)  | 9. (a)  | 10. (d) |
| 11. (c) | 12. (b) | 13. (c) | 14. (d) | 15. (d) | 16. (d) | 17. (d) | 18. (b) | 19. (d) | 20. (b) |
| 21. (d) | 22. (c) | 23. (a) | 24. (b) | 25. (c) | 26. (a) | 27. (b) | 28. (c) | 29. (d) | 30. (b) |
| 31. (d) | 32. (c) | 33. (d) | 34. (b) | 35. (c) | 36. (b) | 37. (b) | 38. (a) | 39. (d) | 40. (b) |
| 41. (d) |         |         |         |         |         |         |         |         |         |

